



Ion irradiation effects on Cr-coated zircaloy-4 surface wettability and pool boiling critical heat flux

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ABSTRACT

The concept of coating the currently used nuclear fuel cladding (zirconium-based alloy, typically Zircaloy-4 or Zirc-4) with an oxidation preventive layer is a progressing Accident tolerant Fuel (ATF) candidate alloys. The coated Zirc-4-based alloys could be a solution to suppress undesirable fast reaction kinetics with high-temperature steam. Zirc-4 has been the most preferred cladding material in pressurized water reactors (PWRs). Chromium (Cr) based alloys as a coating material provides excellent corrosion protection and good strength and wear resistance. This paper presents the surface wettability measurements and pool boiling Critical Heat Flux (CHF) for Cr-coated Zirc-4 claddings pre- and post-exposure to an ion irradiation environment.

The wettability measurements, including static contact angle (contact angle, θ) and average surface roughness (surface roughness, R_a), are introduced for samples of different coating thicknesses (5–30 μm thick). The coatings fabricated by the cold spray of Cr-Al particles to 10 mm $\times$ 10 mm $\times$ 1.95 mm Zirc-4 substrates. Post fabrication, a Pilgering (cold rolling) process, was applied to finalize the coating thickness and resulted in a significant reduction in surface roughness of initially fabricated rough surfaces. The process produced three distinguished samples 5- μm unpolished (as machined), 5- μm , and 30- μm polished (cold rolled). The measurements are presented for the three surfaces and bare Zirc-4 as a baseline surface. The contact angle analyses were implemented in theoretical models from the literature to predict pool boiling CHF. Pool boiling experiments were conducted to measure the pool boiling CHF values and compare them to the predicted values. Scanning Electron Microscope (SEM) images and Energy Dispersive X-ray Spectroscopy (EDS) analysis was performed to characterize the surfaces for better understanding and interpreting the results.

The SEM images showed localized surface damage due to ion irradiation. No recognized change in the measured surface roughness due to ion irradiation. The contact angles of irradiated Cr-coated surfaces are consistently higher (10°) than pre-irradiated surfaces. Decreasing the Cr-coating layer thickness resulted in lower contact angle pre- and post- ion irradiation. The predicted pool boiling CHF using the Kandlikar model is in good agreement with the experimentally measured CHF values within $\pm 12\%$ for all samples.

1. Introduction

The Accident Tolerant Fuel (ATF) program started in early 2011 following the Fukushima Daiichi nuclear power plant accident to initiate the development of new nuclear fuel and cladding materials (Goldner, 2012). These materials are designed to maintain or improve the performance and safety characteristics under both normal and severe accident conditions of Light Water Reactors (LWRs.) (Zinkle et al., 2014). Many desirable features of the ATF candidate alloys, a lower oxidation rate at a higher temperature is a crucial factor (Pint et al., 2015). Zirconium-based alloy (typically Zircaloy-4 or Zirc-4) have been the preferred cladding material in Light Water Reactors (LWRs) for its stable properties. Zirc-4 has low neutron absorption cross-section,

stable heat transfer properties, corrosion resistance, and other preferred mechanical properties (Kim et al., 2016a). Chromium (Cr) based coatings provide excellent corrosion resistance with high melting temperature, and exceptional strength and wear resistance. To some extent, it is believed that Cr-coated Zirc-4-based claddings have the potential to be the fastest route to the ATF.

Coated Zirc-4 has been studied to reduce the oxidation embrittlement of nuclear fuel cladding under Loss of Coolant Accident (LOCA) scenarios. The studied coatings included a ceramic, nitride, and metallic multi-layered with thicknesses varied from a few microns to 10 mm. Results showed that Cr-based coatings exhibited the most promising behavior with an excellent oxidation resistance in steam at high temperature, and good adhesion to the substrate (Brachet et al.,

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2019). Cr-coated Zirc-4 showed excellent oxidation resistance under different pressurized water reactor (PWR) standard pressure and steam temperatures of 500 and 1200 °C (Kim et al., 2015, 2018). The exposure time to steam extended up to 20 days for long-term corrosion testing. The Cr-coated Zirc-4 cladding showed better high-temperature corrosion resistance (negligible weight gains due to oxidation) compared to bare Zirc-4 (Ševeček et al., 2018). A similar experiment conducted by exposing Cr-coated Zirc-4 samples to a flow of water vapor at 1200 °C for 3000 s using the thermos-gravimetric analyzer. Results have shown improved oxidation and corrosion resistance under steam environments compared to uncoated Zirc-4 (Kim et al., 2016b; Brachet et al., 2018). Other Cr-coated Zirc-4 samples were exposed to simulated LOCA conditions by conducting a quenching by flooding testing. Compared to bare Zirc-4 alloys, coated samples showed excellent oxidation resistance under the same testing conditions. The ballooning and rupture temperature of the Cr-coated tubes was higher compared to uncoated ones. The circumferential strain and rupture size were much less than the uncoated cladding. The study recommended additional investigations before considering the cladding candidate materials for practical use (Park et al., 2016; Lee et al., 2019a,b). A cold sprayed Cr coating on Zirc-4 substrates was investigated to support the ATF cladding program. The spray parameters and particles properties were varied to attain the desired coating thickness, microstructure, and oxidation resistance. Samples investigated under a wide range of temperatures in the air and steam environments showed high oxidation resistance. The cold sprayed Cr-coated Zirc-4 demonstrated promising results for enhancing the accident tolerance of zirconium-alloy fuel cladding in LWRs in terms of performance and cost-effectiveness (Maiera et al., 2019).

Evaluation of the thermal performance of ATF candidate materials, including Cr-coated Zirc-4, is progressing. The heat transfer performance, including Critical Heat Flux (CHF) (also known as a departure from nucleate boiling), is safety limits for LWR technology to be evaluated (Brown et al., 2016). The CHF is a phenomenon associated with flow and pool boiling heat transfer at which bubbles coalesce and cover the cooled surface with a vapor blanket. The vapor blanket reduces the local heat transfer and abruptly the surface temperature. The fast temperature gradient induces thermal stress and causes the material to fail, or meltdown would occur (Ali and El-Genk, 2013; Liang and Mudawar, 2018; Kim et al., 2017, 2018; Seo et al., 2016). It is worth mentioning that research efforts are ongoing to explore the transient and steady-state flow boiling CHF on different ATF candidate materials. These efforts would provide more insights and information for the in-pile CHF performance but it is outside the scope of this research (Son et al., 2019; Liu et al., 2017; Brown et al., 2017; Lee et al., 2019a,b).

The pool boiling CHF, which is the focus of this research, has less attention compared to flow boiling CHF. The pool boiling could be the dominant reactor cooling mechanism under the loss of fluid circulation and reactor vessel quenching accident scenarios. Son et al. (2018), investigated the effects of chrome coating thickness on pool boiling CHF. The pool boiling CHF experiments were conducted under atmospheric saturation conditions. Surfaces studied were bare nichrome, two types of chromia, and chrome coated wires. Different coating methods, boiling deposition of chromia nanofluid, and RF and DC-sputtering, were applied. Chromia nanofluid was prepared by 100 nm diameter chromia particle, and distilled water, and results showed that the chromia coating significantly increased CHF by 88.8%. CHF enhancement ratio for chromia coating with RF sputtering was 93.8%, while chrome coating with DC sputtering showed 6.55% CHF enhancement.

The static contact angle was measured for all surfaces and implemented in the Kandlikar model (Kandlikar, 2001) to predict CHF before conducting experiments. The predicted CHF showed higher values compared to the experimentally measured CHF. Kam et al. (2015) experimentally measured the pool boiling CHF of De-Ionized (DI) Water on Cr-coated Zirc-4 of two different coating thicknesses (1 and 10 µm). The Cr-coating was fabricated using electrochemical deposition over a 50 × 15 × 0.55 mm Zirc-4 substrate. Results showed lower measured

CHF for the Cr-coated samples compared to bare Zirc-4 with no recognized thickness effect within the experimental range. The hydrophobic properties (large contact angle) of the coated surface resulted in the poor CHF. The measured contact angle for the Cr-coated surface was 80° compared to 59° for bare Zirc-4. Ali et al. (2017,2018) investigated the oxidation effects on surface wettability and measured pool boiling CHF of DI water on ATF iron-based alloys (FeCrAl). Samples oxidized under simulated LWRs conditions (pressure, temperature, and water chemistry). The surface static contact angle was measured and implemented into the Kandlikar model to predict the pool boiling CHF. The experimentally measured CHF, and the predicted values were in good agreement within the uncertainty of the prediction model. Recently, Jo et al. (2019) performed pool boiling CHF experiments for Cr-coated Zirc-4 candidate ATF cladding. The coatings fabricated using cold spray technique, and coating's thickness varied between 64 and 79 µm. Surfaces were polished to match the currently used cladding surface roughness. The measured contact angle for the Cr-coated Zirc-4 sample was $77^\circ \pm 6^\circ$ compared to $64^\circ \pm 3^\circ$ for bare Zirc-4. The experimentally measured pool boiling CHF was 580 and 630 kW/m² for coated and bare Zirc-4, respectively. The study showed no significant change in CHF due to after fabrication polishing treatment regardless of the coating material.

The exposure of the coating to neutron radiation environments would result in microstructure damages and consequently changing the surface properties of the cladding. The in-pile neutron irradiation is a process associated with severe surface damage in which primary knock-on atoms induced by the energetic ion bombardment with a surface in the crystal lattice (Yan et al., 2015). The irradiation damage could create points and clusters defects that cause changes to the material's physical and mechanical properties (Fluss et al., 2002). The altered surface characteristics would strongly affect surface wettability and CHF (Kim et al., 2013, 2017). Takamasa et al. (2005), investigated the effect of Radiation-Induced Surface Activation (RISA) on surface wettability and CHF. Surface wettability measured on Aluminum, Stainless Steel, and Zirc-4 surfaces exposed to gamma irradiation at different pressure and temperature. Results showed decreased contact angle at a different rate according to the radiation dose, exposure time, temperature and pressure, and the waiting time to collect measurements.

In summary, results indicate that surface wettability (contact angle) has a remarkable effect of on the pool boiling CHF (Phan et al., 2009). The wettability plays a role in the bubble growth and re-establishing the thermal boundary layer by rewetting the surface following the bubble departure from the surface-active nucleation sites (Jeong et al., 2009). The surface wettability is strongly affected by the surface exposure environment that could alter the surface characteristics, including oxidation and irradiation. Surface roughness is another factor that not only affects the surface wettability but also directly changes the wetted surface area for heat transfer. Increased surface roughness would increase the possible active nucleation sites, which would impact the CHF (El-Genk et al., 2013). Neutron irradiation experiments are time-consuming, expensive, and requires specialized facilities to analyze samples with residual radioactivity. Ion irradiation can simulate the effects of in-pile neutron damage for several years in much shorter exposure time. Ion irradiation experiments parameters are well-controlled, the primary atom knock-on energy is highly comparable to neutron irradiation, and the overall process is economical. However, the damage is local within several millimeters or hundreds of nanometers (Was, 2007a,b).

The objective of this research is to present the ion irradiation effects on surface wettability and pool boiling CHF for Cr-coated Zirc-4 ATF candidate alloy. The surface wettability measurements, including contact angles and surface roughness, were collected for Cr-coated Zirc-4 based samples of different coating thicknesses pre-and post-ion irradiation. Surfaces were characterized using a scanning electron microscope (SEM) and energy-dispersive X-ray spectroscopy (EDS) pre- and post-ion radiation. The experimentally measured CHF values are

compared with the predicted CHF using multiple theoretical models in the literature (Zuber, 1959; Kandlikar, 2001; Liao et al., 2008).

2. Experimental facilities and procedures

2.1. Measurement samples

The static contact angle and surface roughness are measured for bare Zirc-4 and three Cr-coated Zirc-4 samples of different coating thicknesses fabricated by the cold spraying technique. The Zirc-4 substrate ($10 \text{ mm} \times 10 \text{ mm} \times 1.95 \text{ mm}$) was cleaned initially in deionized water, acetone, and alcohol solution using an ultrasonic cleaner to remove any organic substances. Grounded 26–45 μm -diameter Cr-Al particles accelerated in N_2 gas at 800 $^\circ\text{C}$ and a pressure of 30 bar through the spray nozzle oriented at a right angle to the substrate located 35 mm apart. The cold spray process resulted in rough-coated surfaces which are not desirable for application in nuclear fuel cladding. Surfaces were smoothed using a cold Pilgering process at room temperature (commonly used for manufacturing claddings). The reduction in the coating thickness was 36%, and a remarkable decrease in the surface roughness was observed after cold rolling with no surface cracking. The final Cr-coating thicknesses after the cold rolling and polishing ranged from 5 to 30 μm . More details on the fabrication process of the Cr-coated Zirc-4 samples are available in the literature (Park et al., 2016).

Two different sample sets of various thicknesses used in the wettability measurements, 5- μm polished (cold rolled), 5- μm unpolished (as fabricated), and 30- μm polished (cold rolled). The first set used for surface wettability measurement pre- and post-ion irradiation and conducting the post-irradiation pool boiling CHF experiments. The second set of samples used for the pre-irradiation wettability measurements and CHF experiments.

2.2. Ion irradiation

The Cr-coated ion irradiation was performed at Sandia National Laboratories (SNL) Ion Beam Laboratory using a 5 MeV Al^+ beam produced by a 6 MV Tandem accelerator. 5 MeV Al^+ was selected to eliminate any chemical interaction with the coating elements (Cr and Al) and generate damage within the thickness of the coating layer only. The ion irradiation conducted at room temperature to a fluence of 2.6×10^{16} ions/ cm^2 . The experimental parameters were simulated using the Stopping and Range of Ions in Matter (SRIM) software to determine the damage associated with the irradiation in the atomic displacement per atom (dpa) unit (Ziegler et al., 2010; Stoller et al., 2013). Fig. 1 shows that all surface damage due to ion irradiation occurred in the first 2.5 μm with an average rate of 2.47×10^{-5} dpa/s.

2.3. Contact angle

The contact angle and surface roughness were measured using a general Goniometer and Dektak surface profilometer instruments, respectively. The contact angle goniometer measures contact angle, surface tension, and surface energy. The facility has an automated dispensing system to control the droplet volume and ensure measurement repeatability. The controlled tilting base can change the specimen tilting angle to measure the advance and receding contact angles, respectively. The machine has a temperature-controlled chamber for up to 300 $^\circ\text{C}$ measurements.

The American Society of Testing Materials (ASTM) published a fully detailed standard procedure for measuring the surface contact angle using Goniometer (ASTM D5725, 2013; D7334, 2013). The document summarized the measurement of the contact angle of liquid droplets on metal substrates. The ASTM standard provides the necessary number of droplets, droplet volume, and measuring time to ensure accurate measurements. According to the procedure, the contact angle of a

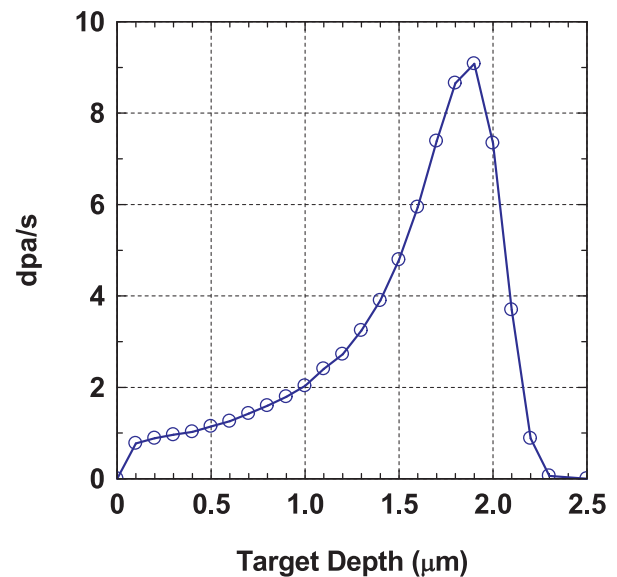


Fig. 1. Displacement production in Cr-coated samples irradiated with 5 MeV Al ions.

surface is the average of three contact angle measurements of individual droplets with a volume of approximately 20 μL each. The measuring time from the moment of attaching the droplet to the surface should not exceed 30 s. The short measuring time is to ensure the volume of the droplet does not change by evaporation or diffusion into the substrate. In this work, the contact angles measured following the ASTM procedure, but with more droplets to minimize the uncertainty in the representative contact angle value. The number of droplets was enough to cover the full surface area of individual samples (four droplets per Cr-coated surfaces and eight droplets for bare Zirc surface). The contact angle was recorded 60 times within 30 s for each Deionized (DI) water droplet using 16–20 μL droplets' volume. The 60 readings were averaged to present a single measurement for each droplet. The variation between the 60 values and the average was within ± 1 –1.5% for all droplets.

2.4. Surface roughness

The surface profilometer (Dektak 150) is a stylus-type with 1 mg force to measure surface peaks and valleys. The average surface roughness, R_a , of each scan is calculated from the surface topology measurements using the Dektak software. The machine can measure samples up to 12.7 cm in diameter. The system has a 1-millimeter standard vertical range with up to 120,000 data points per scan. The Dektak 150 stylus profiler is a device used to measure the hills and valleys of the surface topology and calculate the average surface roughness. The average surface roughness values, R_a , presented is the arithmetic mean of the roughness measurements along with each scan computed by the Dektak following the equation, $R_a = \frac{1}{n} \sum_{i=1}^n |y_i|$.

Four standard scans each 8 mm long with a 1.33 $\mu\text{m/s}$ scan rate recorded to calculate the average roughness, R_a for each surface. Fig. 2 presents a sample of three individual scans raw data carried out on unirradiated bare Zirc-4, and the individual calculated R_a values (Fig. 2a–2c). Fig. 2d compares the individual scans with the estimated overall R_a value for the combined scan raw data. The deviation between the total R_a value (0.35 μm) and the different scans R_a (0.337, 0.36, and 0.351 μm) is within ± 5 –7% for all specimens. Thus, using the arithmetic average of the individual scans raw data is accurate enough to present the R_a value for each sample, as discussed in the results section.

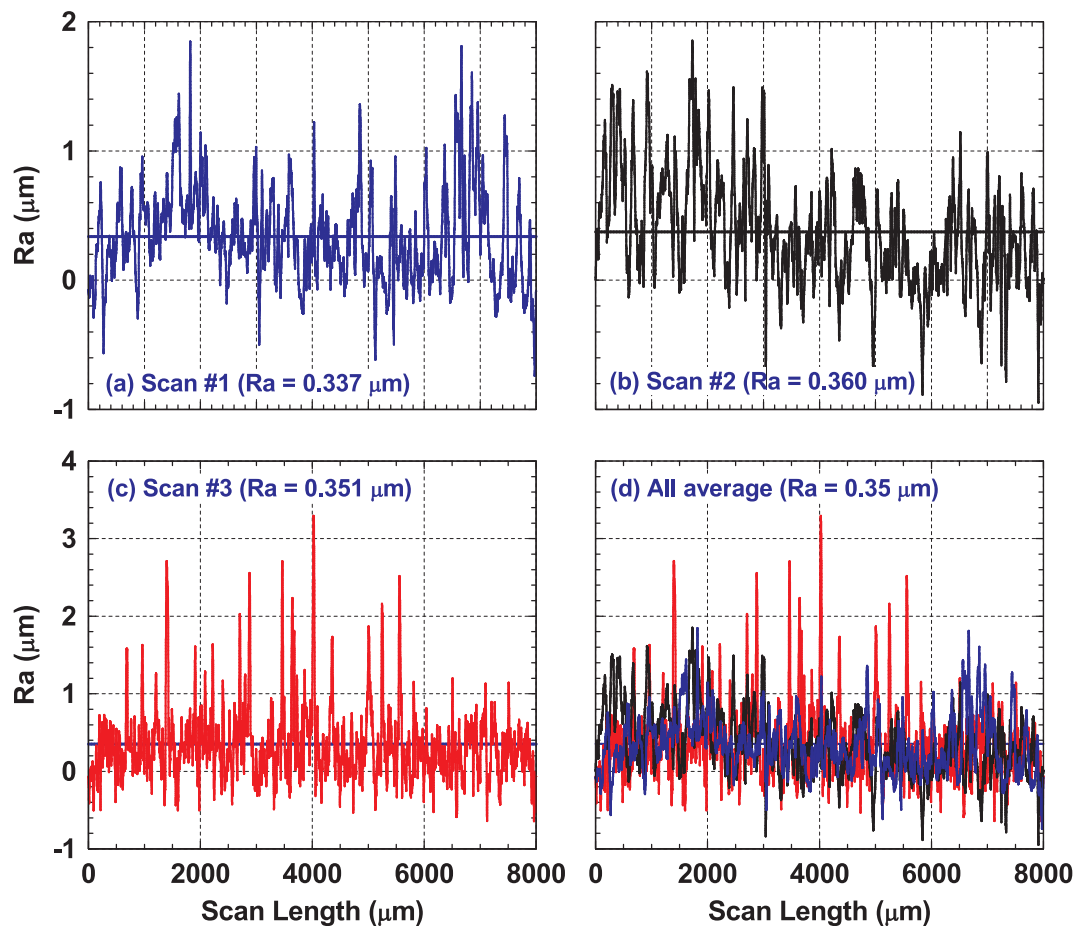


Fig. 2. Different scans (a-c) and overall average (d) Dektak calculated Ra values.

2.5. Pool boiling

The pool boiling experimental facility (Fig. 3) composed of an $80 \times 150 \times 230$ mm stainless steel boiling chamber (1) with front and rear tempered glass windows for visualizing the process (2). The facility is fitted with two submerged heaters 400 W each (3) for initial water heating and liquid degassing. Two 200 W surface heaters are attached to the chamber sides (not shown) to keep constant water temperature during the experiments. The chamber lid is fitted with a surface condenser (4) to condense the generated vapor and keep the water level

and chamber pressure constant during the experiment. The chamber cover has two feed-through holes (5) for thermocouples to measure water and vapor space temperatures. The test section assembly (6) is made of poly-ether-ether-ketone (PEEK) and installed through the bottom of the chamber.

The test surface was connected to a copper block using high thermal conductive epoxy (Circuit-works), and both attached to the heating element with the same epoxy. The assembled heated section inserted into the assembly cap and two-component epoxy (Omega-Bond) applied to insulates the boiling surface sides. The epoxy insulation

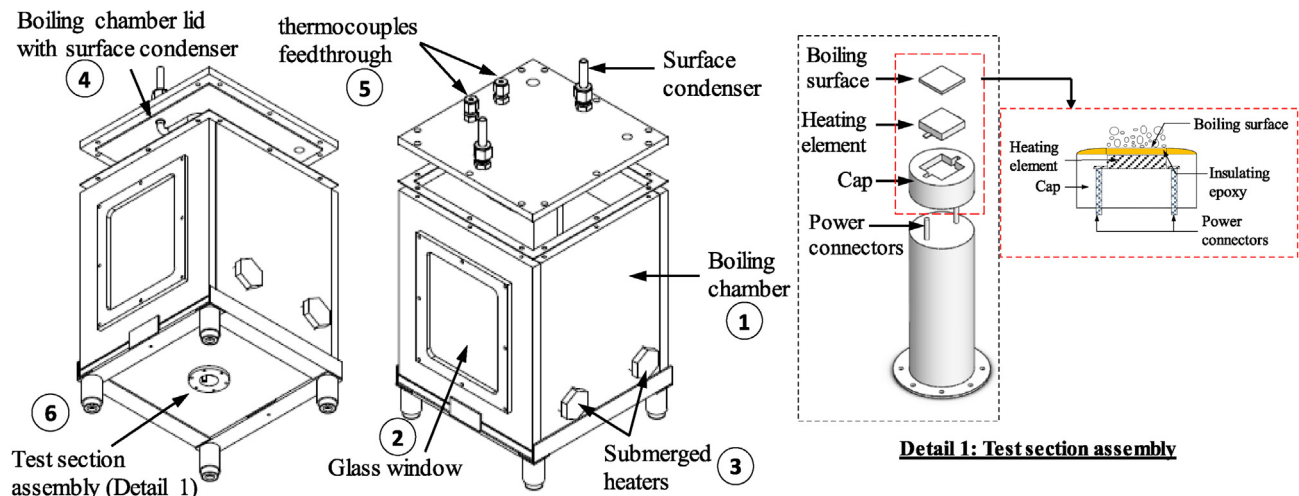


Fig. 3. Schematic of the pool boiling facility.

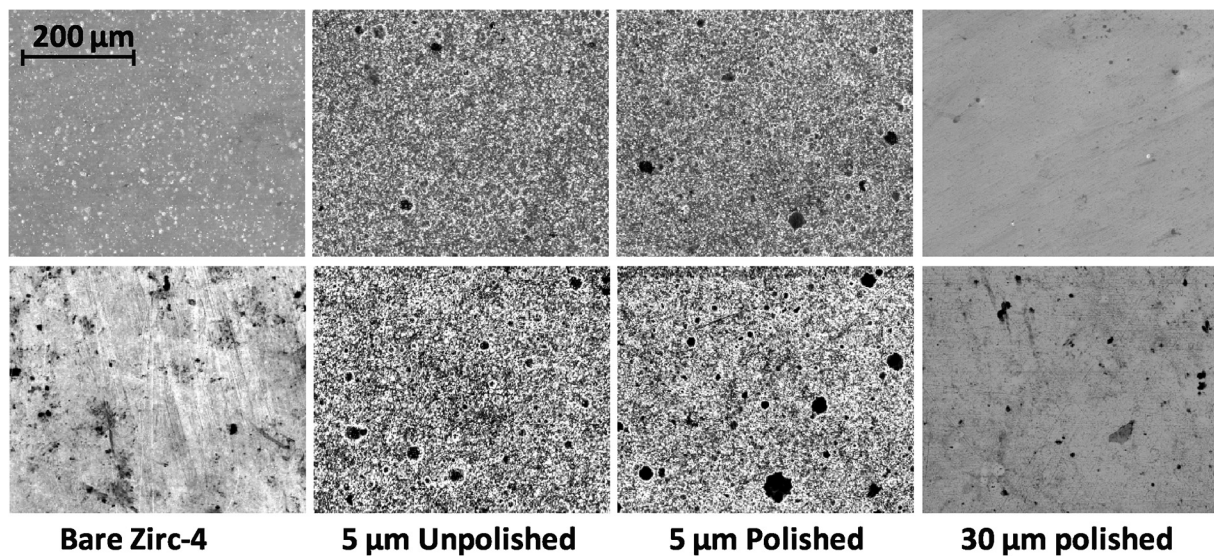


Fig. 4. SEM images pre (top row)-and post (lower row)-ion irradiation.

ensured the heat dissipated by the heating element only removed by nucleate boiling only from the top face of the tested surface.

The two small holes in the cap allow connectors through the PEEK column to supply power to the heating element. The copper block temperature measured with multiple K-type thermocouples (TCs) impeded into the block. The copper block temperature used to calculate the heated surface temperature using a linear conductive heat transfer model. The bulk fluid temperature also recorded by two different TCs. Measurement Computing DAQ system controlled and collected the voltage, current, and temperature readings through the LabView interface. The nominal uncertainty of the TCs is ± 1.1 °C, according to the manufacturer, OMEGA Engineering. The experimental uncertainty of the CHF was calculated to be 1.042%.

3. Surface characterization

Before collecting measurements or conducting experiments, surfaces were characterized using SEM and EDS analysis to track any surface irradiation damage and interpret any measurement changes. Fig. 4 shows the SEM images and the local surface damage induced by the bombardment of Al⁺ on all samples. The local defects shown in the SEM images would impact the surface roughness and contact angle measurements, and directly change the predicted and experimentally measured pool boiling CHF. The EDS analysis provided the element weight percentage (wt.%) for the Cr-coated Zirc-4 samples pre- and post-ion irradiation is presented in Fig. 5. The figure clearly shows that the Al signal increased in the coated samples post-irradiation because of the Al⁺ implantation.

4. Measurement results

4.1. Contact angle

The measured contact angles using four droplets on the Cr-coated and eight droplets on bare Zirc-4 samples (four for each face), were measured, and values are presented in Figs. 6 and 7. These figures compare the individual droplets and overall average contact angle for different surfaces. Fig. 6 shows that the 30-μm thick Cr-coated sample has the lowest surface wettability (higher average contact angle, 73.49°). Also, the figure indicates that the 5-μm thick polished Cr-coated surface has the highest wettability (smallest average contact angle, 68.4°). The measured contact angles for bare Zirc-4 and 5-μm thick Cr-coated surfaces are 71.82°, and 68.87° respectively. Fig. 7

presents the contact angle measurements for the same samples post-ion irradiation. The measured contact angles for all surfaces have shown a consistent increase in wettability, preserving the order of samples. The post-irradiation measured contact angle values are 83.39°, 81.2°, 79.46°, and 79.31° for 30-μm thick polished, bare Zirc-4, 5-μm thick unpolished, and 5-μm polished Cr-coated samples, respectively.

4.2. Surface roughness

Scan routes were selected to cover most of the sample's surface area and avoid any surface scratches. Fig. 8 (left) shows the Dektak calculated four distinct surface roughness measurements and the corresponding overall averages for pre-irradiation samples. The measured Ra for 5-μm thick unpolished Cr-coated is the highest (0.85 μm). The average Ra values are 0.66, 0.35, and 0.12 μm for 5-μm polished, bare Zirc-4, and 30-μm thick polished Cr-coated samples. The irradiated surface roughness is presented in Fig. 8 (right). The average Ra values are 0.87, 0.66, 0.34, and 0.13 μm for 5-μm unpolished, 5-μm polished, bare Zirc-4, and 30-μm thick polished irradiated samples, respectively.

4.3. Pool boiling CHF

Fig. 9 shows the pool boiling CHF and heat transfer coefficient results for Cr-coated and bare Zirc-4 pre- and post-irradiation selected surfaces (bare and 30-μm Cr-coated Zirc-4). Although, the figure is not presenting a clear order of the heat transfer curves for pre- and post-ion irradiation samples, but it shows a higher CHF of pre-irradiated over post-irradiated surfaces.

The results in the figure show that the CHF for bare Zirc-4 is 784.14 and 749.66 kW/m² for pre- and post-ion irradiated samples, respectively. These CHF values decreased to 711.67 and 676.83 kW/m² for pre- and post-ion irradiated 30 μm thick Cr-coated Zirc-4 samples, respectively. It is worth to mention that, the limited number of Cr-coated Zirc-4 samples limited the number of experiments to check the reproducibility of the data except for the bare Zirc-4 sample. Three separate tests conducted, and the CHF values were within the experimental uncertainty (1.042%).

5. Discussion

Fig. 10 compares the surface roughness and contact angle measurements pre- and post-ion irradiation. The results indicate almost no effects of the ion irradiation on the measured surface roughness. The

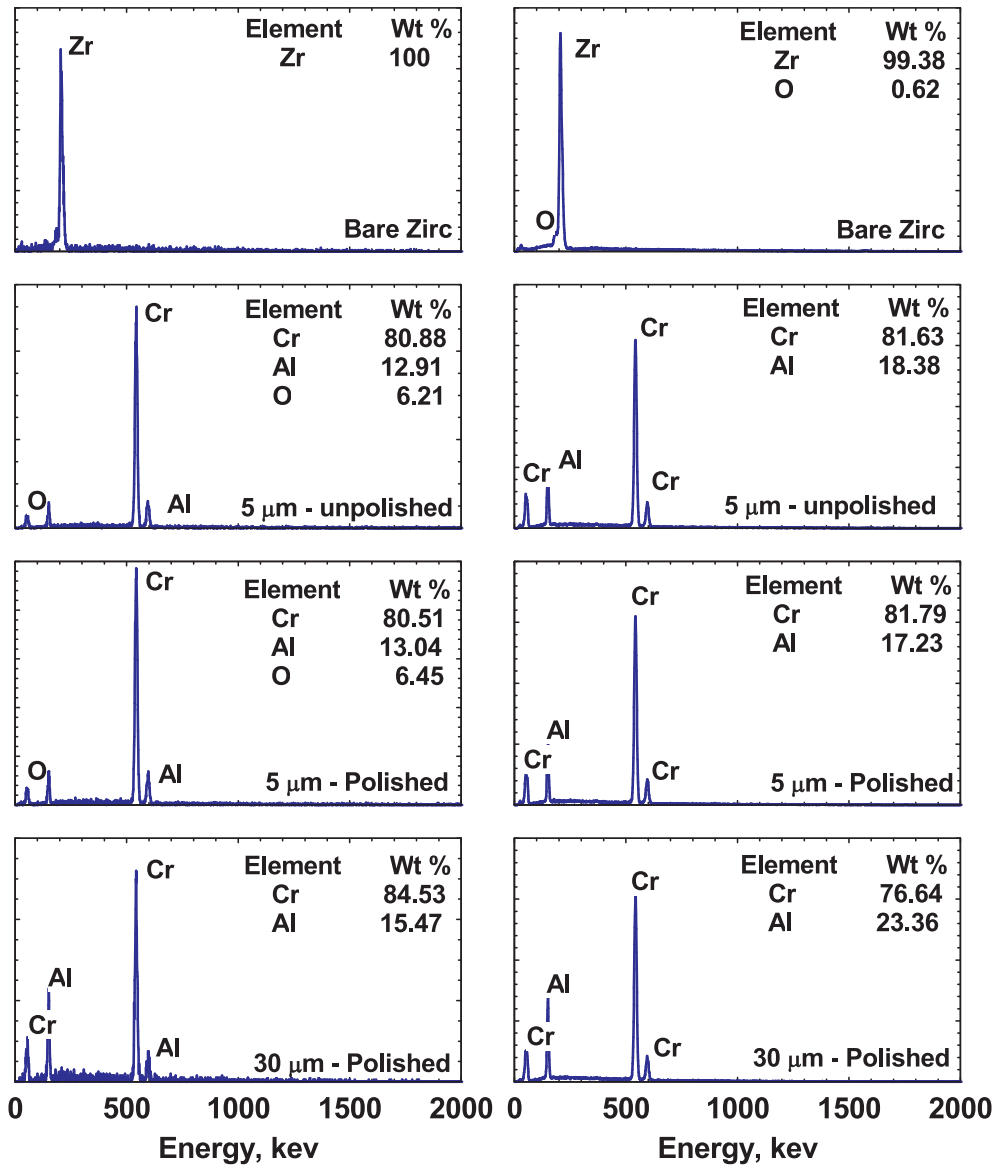


Fig. 5. EDX elemental analysis of Cr-coated Zirc-4 based samples pre- (left)- and post (right)-ion irradiation.

surface damages on the irradiated surfaces are local and small to have a significant impact on the measured surface roughness (Fig. 10a). Fig. 10b presents the comparison of the contact angle measurements pre- and post-ion irradiation. Results show a consistent increase by an average of 10° in the measured contact angles for all irradiated samples. The contact angle is the equilibrium angle of a water droplet on a solid surface, which is directly related to the interfacial energy balance between different phases. The consistent increase in the measured contact angle indicates that the ion irradiation changes the surface energy of the solid surface because of the microstructure defects and presence of the bombarding ions. The surface energy change due to ion irradiation was confirmed by other researchers, especially when Cr-element was present (Eremin et al., 2017).

Researchers recognized the impact of contact angle on CHF, and models were developed to predict the pool boiling CHF and reported in the literature (Zuber, 1959; Kandlikar, 2001; Liao et al., 2008). These models compared to many experimental CHF data on surfaces of different degrees of wettability for validation. Zuber (1959), developed an equation to predict the CHF from the hydrodynamic instability model without accounting for surface characteristics. The model was a function of the thermophysical properties of the boiling liquid.

$$q_{CHF} = Ch_{lg}\rho_g^{0.5}[\sigma g(\rho_l - \rho_g)]^{0.25} \quad (1)$$

Kandlikar (2001) developed a correlation to predict the CHF in saturated pool boiling, considering the impact of surface wettability. A model accounting for receding contact angle, surface orientation, and sub-cooling effects were developed. The model was validated by comparing the predicted CHF values with a wide range of experimental data from the literature. The model agreed with experimental data within an error of $\pm 21.59\%$:

$$q_{CHF} = h_{lg}\rho_g^{0.5}[\sigma g(\rho_l - \rho_g)]^{0.25} \left(\frac{1 + \cos\theta_r}{16} \right) \left[\frac{2}{\pi} + \frac{4}{\pi}(1 + \cos\theta_r)\cos\phi \right]^{0.5} \quad (2)$$

Liao et al. (2008), fitted a correlation for CHF taking the effects of both orientation and static contact angle into account (the equation is presented for upward facing, $\phi = 0^\circ$):

$$q_{CHF} = h_{lg}\rho_g^{0.5} \left[1 + \frac{55 - \theta}{100}(0.56) \right] [\sigma g(\rho_l - \rho_g)]^{0.25} \quad (3)$$

Where, C is a constant, h_{lg} is the liquid latent heat, ρ_l and ρ_g are the liquid and gas density of water. The σ is the liquid surface tension, θ_r

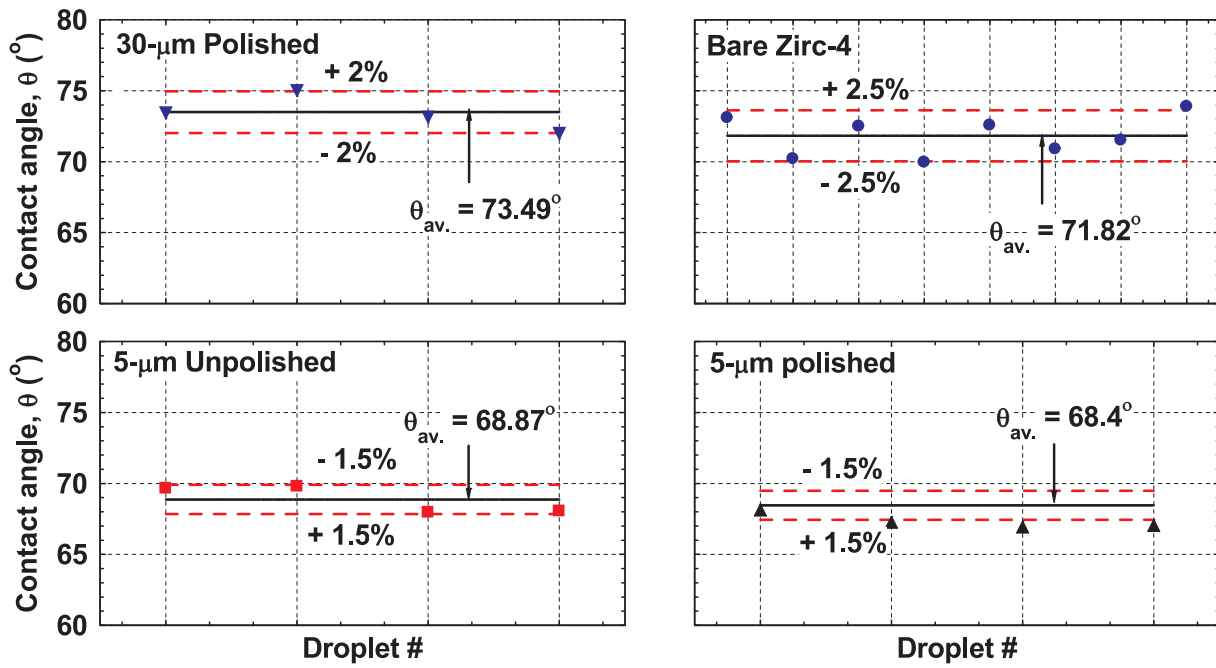


Fig. 6. Measurements of static contact angle on pre-irradiated samples.

and θ are the receding and static contact angles respectively, and ϕ is the surface inclination angle.

The measured contact angles (Fig. 10b) were implemented in the models listed above (Eqs. 1–3) to predict the pool boiling CHF of Cr-coated Zirc-4 pre- and post-ion irradiation, as shown in Fig. 11. Zuber correlation (Eq. (1)) predicts equal pool boiling CHF of 1027 kW/m² regardless of the contact angle changes due to irradiation effects. Other models (Eqs. (2) and (3)) predicted higher CHF for Cr-coated samples pre-irradiation compared to irradiated samples. The pool boiling CHF ranges 808–878 kW/m² and 673–728 kW/m² for pre- and post-irradiated samples using the Kandlikar model (Eq. (2)).

In general, the higher surface roughness indicates a larger wetted surface area in contact with the liquid compared to the footprint area.

The increased wetted area would increase the number of surface microcavities and hence possibly increase the active nucleation sites density. Increased active sites for bubble nucleation and the wetted surface area increases the boiling heat transfer coefficient and CHF. The high wettability promotes the rewetting of the dry spots developed on the surface immediately after the bubble departure from the surface and delays the occurrence of CHF. Based on the obtained results, the 5 μ m thick Cr-coated surfaces expected to provide the highest, and the 30 μ m thick Cr-coated sample would experience the lowest, pool boiling CHF.

Fig. 12 presents a comparison between the predicted CHF using the Kandlikar model and experimentally measured CHF for pre- and post-ion irradiated samples. As shown in Fig. 12, the experimentally measured CHF has consistent results with the CHF prediction. The measured

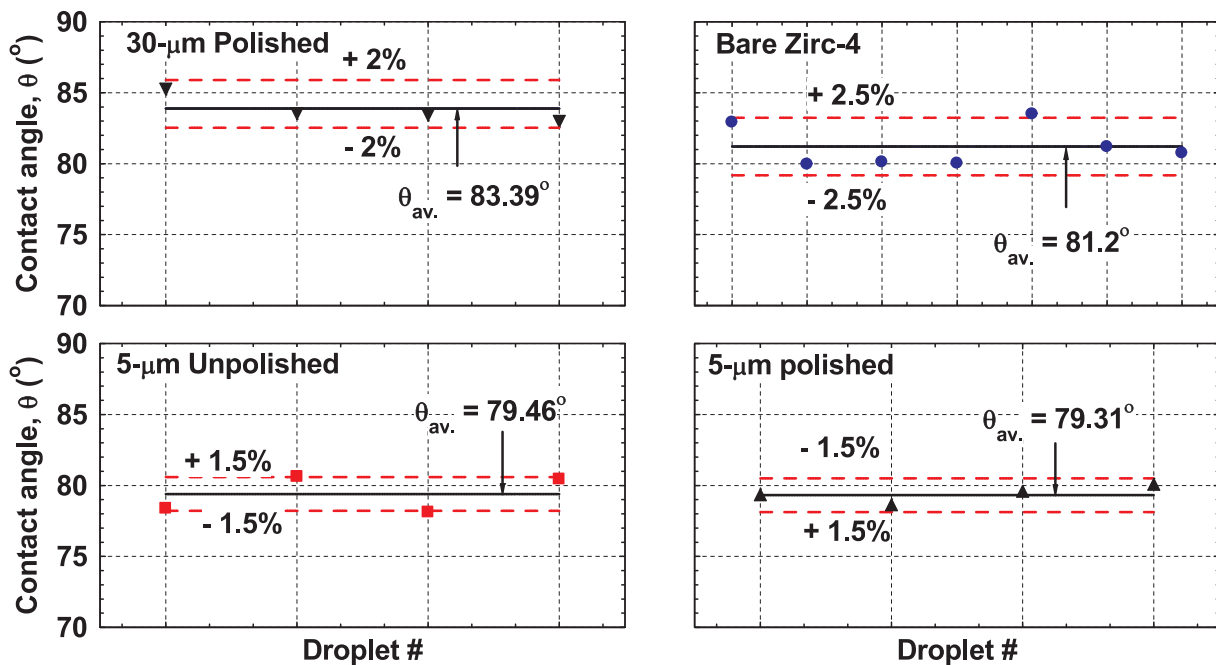


Fig. 7. Measurements of static contact angle on post-irradiated samples.

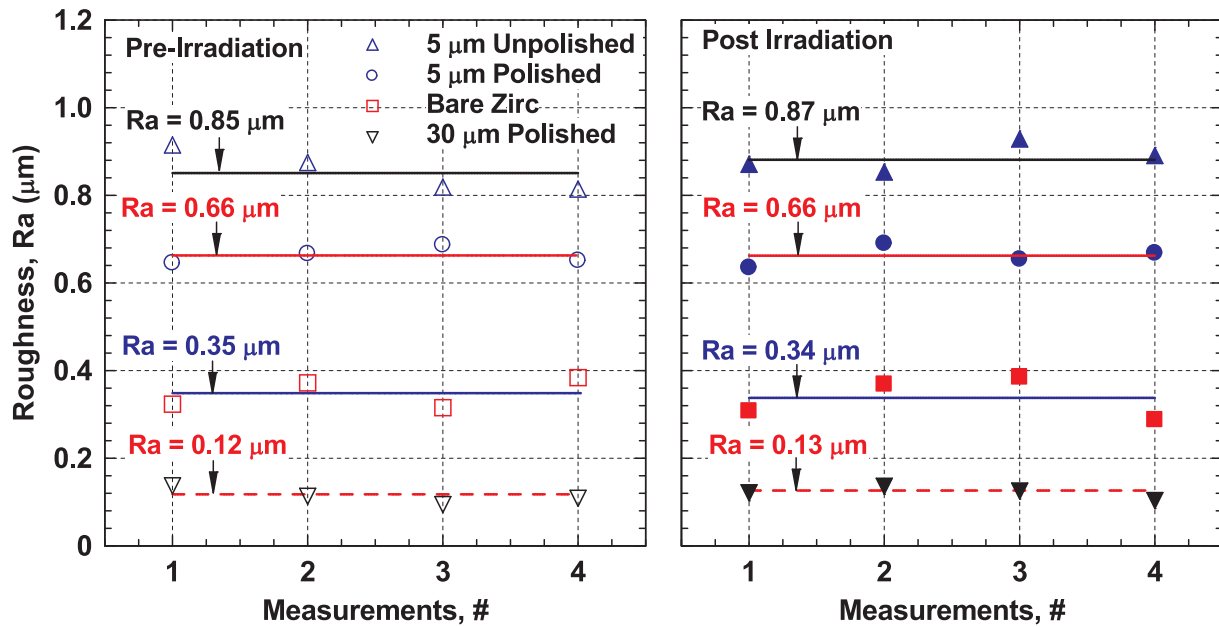


Fig. 8. Measurement of surface roughness, Ra, on pre-irradiated (left) and post-irradiation (right) samples.

CHF is higher for pre-irradiation compared to post-irradiation samples. The higher CHF would attribute to the higher wettability (low contact angle) of the pre-irradiated samples. The CHF values are 815, 790, and 735 kW/m^2 for pre-irradiated 5- μm Cr-coated, bare Zirc-4, and 30- μm thick Cr-coated samples, respectively. The CHF values decreased to 792, 784, and 609 kW/m^2 for the post-irradiated 5- μm Cr-coated, bare Zirc-4, and 30- μm thick Cr-coated samples respectively. The experimentally measured CHF values presented in Fig. 13 are within $\pm 12\%$ of the predicted CHF values using the Kandlikar model. The model used a large pool of experimental CHF data from the literature for validation, and the reported model uncertainty is $\pm 21.59\%$ (Kandlikar, 2001). These results increased the confidence in the obtained experimental measurements. The agreement with the Kandlikar model makes it applicable to implement in a further simulation activity where pool boiling CHF correlation for Cr-coated Zirc-4 cladding is needed.

The pool boiling CHF results for pre-irradiated Cr-coated Zirc-4

surfaces are compared to other data available in the literature for coatings fabricated with different techniques (Fig. 13). The figure clearly shows highly scattered results between surfaces based on the manufacturing method. The measured CHF on surfaces fabricated by nanofluid deposition and RF sputtering shows the highest values. The Cr-coated Zirc-4 samples manufactured by cold spray show very close CHF value. These results indicate that coating thickness and post-fabrication surface finish (polishing or Pilgering) has an insignificant impact on pool boiling CHF compared to the fabrication technique.

6. Summary and conclusion

This study investigated the ion irradiation effects on surface wettability and pool boiling CHF for Cr-coated Zirc-4. The coatings deposited using the cold spray technique. Pilger process applied following the cold spray to finalize the coating thickness. As a result, the

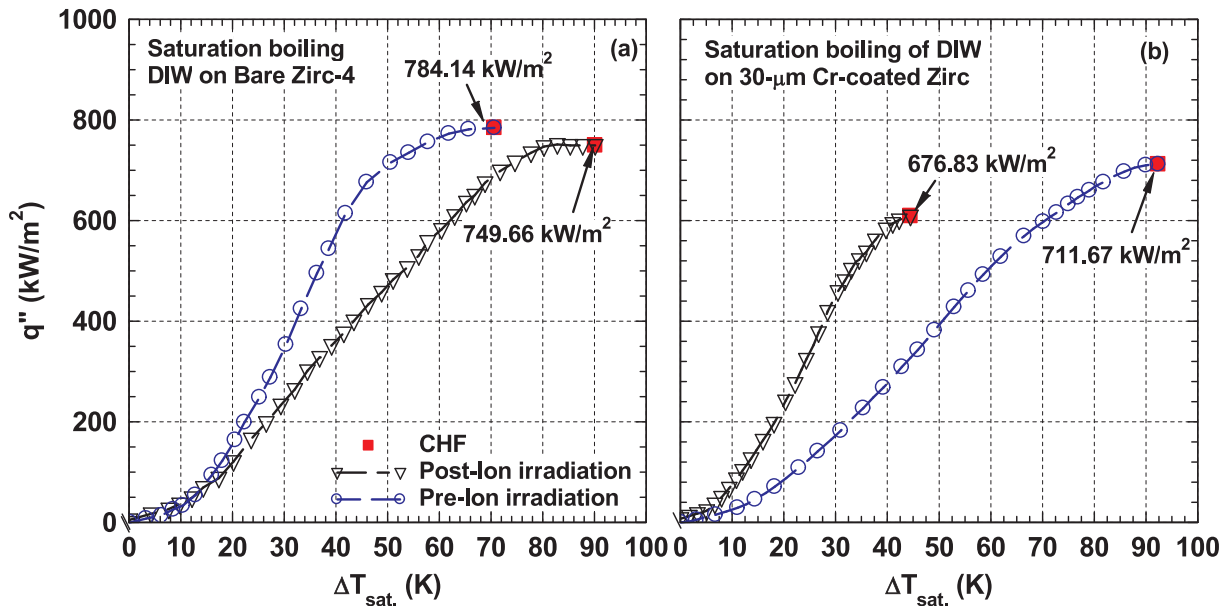


Fig. 9. Pool boiling curves for (a) bare and (b) 30-mm Cr-coated Zirc-4 samples.

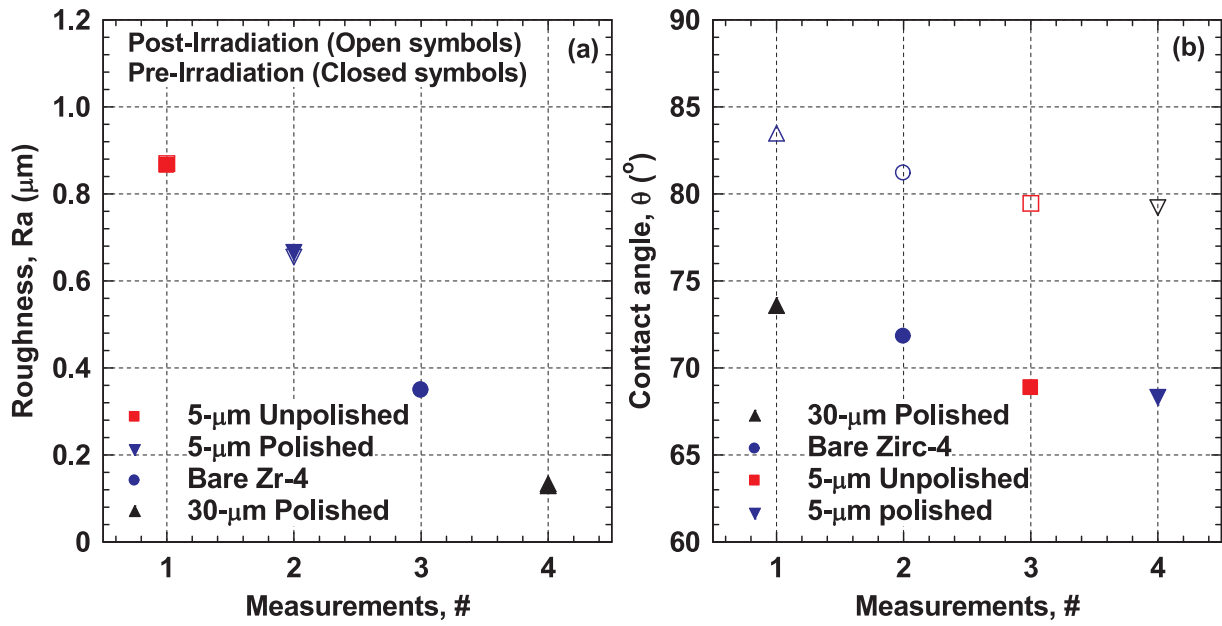


Fig. 10. Ion irradiation effects on the measured (a) surface roughness (b) contact angles.

roughness of the layers significantly reduced. Three distinguished samples were tested 5 μm polished (cold rolled), 5 μm unpolished (as fabricated), and 30 μm (cold rolled) in addition to bare Zirc-4 as a baseline. Samples were exposed to ion irradiation using Al⁺ at an energy level of 5 MeV. The energy and type of ion were selected to eliminate any chemical interaction with the coating elements (Cr and Al) and generate damage within the thickness of the coating layer only. The static contact angle and surface roughness measured pre- and post-ion irradiation. The contact angle implemented in the Kandlikar model and pool boiling CHF was predicted. Pool boiling experiments were conducted to measure the CHF to compare with the model prediction.

The measured static contact angle post-ion irradiation showed a consistent increase by 10° compared to the contact angle measured for pre-irradiation surfaces. The ion irradiation changed the surface energy as a result of the microstructure defects and presence of the bombarding ions, which was confirmed by similar efforts reported in the literature

(Eremin et al., 2017). Other reported results showed a decrease in the measured contact angle post- gamma irradiation (Takamasa et al., 2005). Other findings showed instantaneous decrease immediately post-gamma radiation, and 8 h later, the contact angle increased significantly (Takamasa et al., 2004). These different results indicate the need for additional research efforts to understand the impact of varying irradiation types on the behavior of the post-irradiation contact angle.

The ion irradiation showed no effects on the measured surface roughness pre- and post-ion irradiation. The measured contact angle values were implemented in the Kandlikar model and predicted the pool boiling CHF. The predicted CHF and experimentally measured CHF showed a good agreement within the uncertainty of the model. The measured pool boiling CHF for pre-irradiated Cr-coated Zirc-4 was consistent with the reported CHF in the literature for surfaces fabricated by the cold spray technique (Fig. 13).

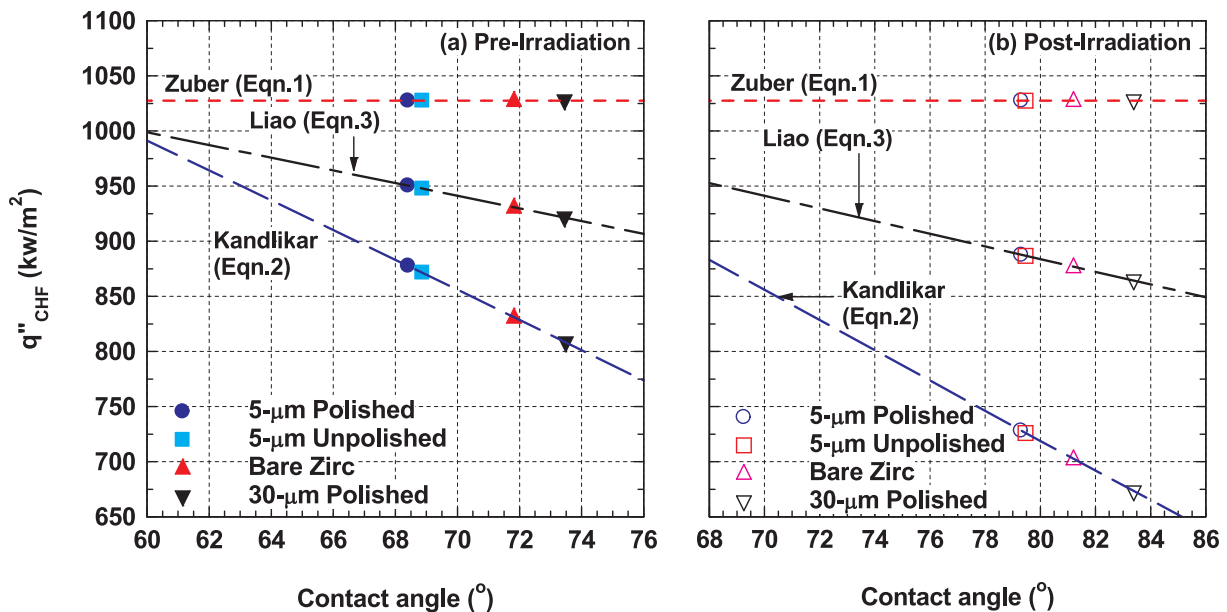


Fig. 11. Predicted CHF based on contact angle measurements.

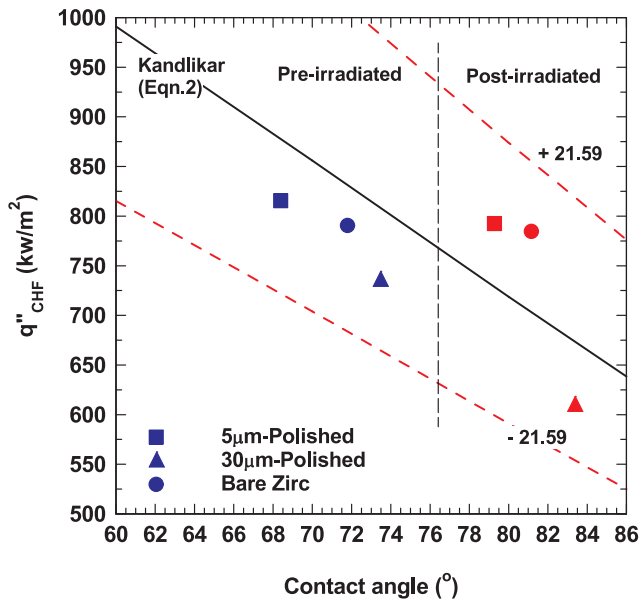


Fig. 12. Comparison between measured and predicted CHF.

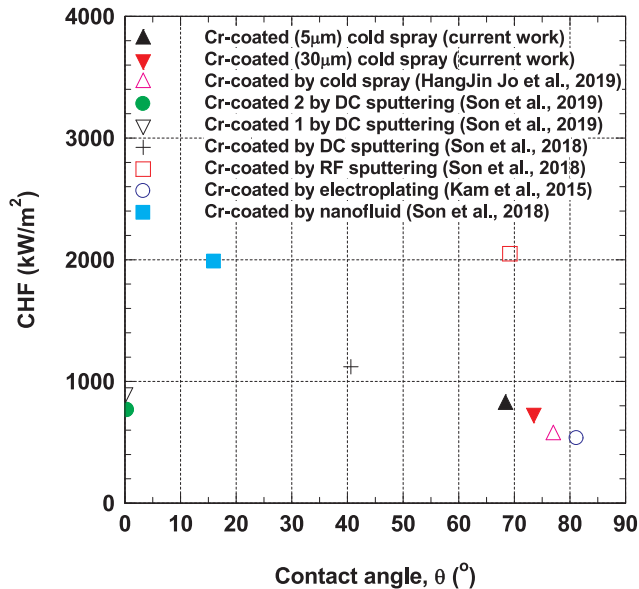


Fig. 13. Comparison between measured pool boiling CHF (pre-irradiation) with literature CHF data for Cr-coated Zirc-4 surfaces fabricated by different techniques.

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